

Intangible Investment Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Intangible Investment Restraint (IIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IIR achieves an annualized gross (net) Sharpe ratio of 0.43 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (19) bps/month with a t-statistic of 2.85 (2.49), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, Accruals) is 19 bps/month with a t-statistic of 2.42.

1 Introduction

Standard consumption-based asset pricing models predict that assets whose payoffs covary positively with consumption growth should earn higher expected returns, as investors demand compensation for bearing systematic consumption risk. Yet identifying firm characteristics that reliably capture cross-sectional variation in consumption risk exposure remains a central challenge in empirical finance. While numerous studies document predictable patterns in stock returns based on accounting variables and investment behavior, the economic mechanisms linking these characteristics to fundamental consumption risk often remain opaque.

We examine whether firms' intangible investment restraint—measured as the negative change in intangible assets scaled by total assets—predicts cross-sectional variation in stock returns through its connection to consumption risk exposure. Firms that exhibit greater restraint in intangible investment may face binding financial constraints or operate in states where the marginal utility of consumption is particularly high, making their cash flows more sensitive to aggregate consumption shocks. This paper tests whether intangible investment restraint captures meaningful variation in firms' exposure to systematic consumption risk and whether this exposure is priced in the cross-section of equity returns.

The consumption-based asset pricing framework of [Lucas Jr \(1978\)](#) and [Breedon \(1979\)](#) establishes that expected returns should compensate investors for bearing covariance risk with the stochastic discount factor, which equals the intertemporal marginal rate of substitution of consumption. Firms exhibiting high intangible investment restraint likely face binding constraints during periods when the marginal utility of consumption is elevated, making their returns particularly sensitive to consumption growth shocks. Building on the long-run risks model of [Bansal and Yaron \(2004\)](#), we hypothesize that firms with high intangible investment restraint load more heavily on persistent components of consumption growth, as these firms' con-

strained investment policies make them more vulnerable to low-frequency economic fluctuations.

The state-dependent nature of intangible investment decisions provides additional insight into consumption risk exposure. During economic downturns when aggregate consumption growth is low and marginal utility is high, firms that previously exhibited investment restraint may experience particularly severe cash flow shocks. This mechanism aligns with the habit formation models of [Campbell and Cochrane \(1999\)](#) and [Constantinides \(1990\)](#), where time-varying risk aversion amplifies the pricing of consumption risk during bad states. Firms with high intangible investment restraint thus carry greater exposure to consumption disasters, as modeled by [Barro \(2006\)](#) and [Gabaix \(2012\)](#), because their limited investment flexibility prevents them from adjusting to severe negative consumption shocks.

The pricing implications follow directly from the first-order conditions of the representative investor’s optimization problem. Assets that covary positively with the pricing kernel—equivalently, those that perform poorly when marginal utility is high—must offer higher expected returns in equilibrium. If intangible investment restraint signals greater exposure to consumption risk, particularly during bad states when risk premia are elevated, then firms with high restraint should earn higher average returns. This prediction emerges naturally from consumption-based models with recursive preferences ([Epstein and Zin, 1989](#)) or rare disasters ([Wachter, 2013](#)), where investors demand substantial compensation for bearing tail consumption risk.

Our empirical analysis reveals that intangible investment restraint strongly predicts cross-sectional variation in expected returns. A value-weighted long-short portfolio that buys stocks with high intangible investment restraint and sells stocks with low restraint generates an average monthly return of 23 basis points with a t-statistic of 3.05, corresponding to an annualized Sharpe ratio of 0.43. The strategy’s alpha relative to the Fama-French six-factor model equals 22 basis points per month with

a t-statistic of 2.85, indicating that standard risk factors cannot fully explain the return predictability associated with intangible investment restraint.

The economic magnitude of the intangible investment restraint premium remains substantial after accounting for transaction costs. The net Sharpe ratio of 0.35 places the strategy in the 93rd percentile among 212 documented anomalies, while the net six-factor alpha of 19 basis points per month retains statistical significance with a t-statistic of 2.49. Notably, the strategy performs particularly well among large-capitalization stocks, generating an average return of 30 basis points per month with a t-statistic of 2.74 within the largest size quintile, suggesting that the effect is not driven by illiquid small stocks that are difficult to trade.

Controlling for closely related investment and financing anomalies barely attenuates the predictive power of intangible investment restraint. When we simultaneously control for six related predictors—including asset growth, accruals, and various financing measures—plus the Fama-French six factors, the strategy alpha equals 19 basis points per month with a t-statistic of 2.42. The loading on the investment factor CMA of 0.24 with a t-statistic of 4.51 suggests that intangible investment restraint captures a distinct dimension of firms’ investment behavior that relates to their systematic risk exposure. These results demonstrate that intangible investment restraint contains unique information about expected returns beyond existing factor models and related anomalies.

This paper contributes to the consumption-based asset pricing literature by identifying intangible investment restraint as a powerful predictor of cross-sectional variation in consumption risk exposure. While [Fama and French \(2015\)](#) and [Hou et al. \(2015\)](#) document that investment-based factors help explain average returns, our results suggest that intangible investment restraint captures a distinct dimension of systematic risk tied to consumption dynamics. Unlike the investment factor in [Fama and French \(2015\)](#), which focuses on tangible investment, our measure specifically

isolates firms' restraint in intangible capital formation, which may be particularly sensitive to long-run consumption risks given the irreversible nature of intangible investments.

Our findings extend the literature on investment-based anomalies, including [Cooper et al. \(2008\)](#) on asset growth and [Titman et al. \(2004\)](#) on capital investments, by demonstrating that intangible investment restraint contains incremental predictive power even after controlling for these related effects. The persistence of the intangible investment restraint premium among large stocks and after transaction costs distinguishes it from many documented anomalies that primarily manifest among small, illiquid stocks. This suggests that the consumption risk mechanism we identify represents an economically meaningful source of systematic risk rather than a market friction or limits-to-arbitrage phenomenon.

The robust performance of intangible investment restraint across different market conditions and its connection to state-dependent consumption risk provides new evidence supporting consumption-based theories of asset pricing. By linking cross-sectional return predictability to fundamental consumption risk through the lens of firms' intangible investment decisions, we offer a theoretically grounded explanation for why certain firm characteristics predict returns. These results have important implications for portfolio construction and risk management, as they identify a previously unexplored dimension of systematic risk that commands a significant premium in equity markets and expands the investable opportunity set beyond traditional factor models.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies.

Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in investment activities and intangible asset positions. Intangible Investment Restraint signal relies on two key accounting variables. Increase in investments (IVCH) represents the cash outflow for acquisitions and investments during the fiscal year, capturing firms' expenditures on acquiring other businesses, purchasing investment securities, and making other long-term investments. This item reflects the net cash used for investment activities beyond regular capital expenditures. Intangible assets (INTAN) measures the book value of a firm's intangible assets net of accumulated amortization, including items such as goodwill, patents, trademarks, copyrights, and other non-physical assets recognized on the balance sheet. We construct the Intangible Investment Restraint signal by calculating the year-over-year change in investment increases, scaled by lagged intangible assets. Specifically, we compute $(IVCH(t) - IVCH(t-1)) / INTAN(t-1)$, where t denotes the current fiscal year. This ratio captures the acceleration or deceleration in investment activity relative to the firm's existing intangible asset base. A positive value indicates increasing investment momentum, while a negative value suggests restraint in investment growth. We calculate this measure using fiscal year-end values and require non-missing observations for both IVCH and INTAN variables. To ensure meaningful scaling, we exclude firm-years where lagged intangible assets equal zero or are negative. The resulting signal provides a standardized measure of how firms adjust their investment pace relative to their accumulated intangible capital.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IIR signal. Panel A plots the time-series of the mean, median, and interquartile range for IIR. On average, the cross-sectional

mean (median) IIR is -0.39 (0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input IIR data. The signal’s interquartile range spans -0.01 to 0.01. Panel B of Figure 1 plots the time-series of the coverage of the IIR signal for the CRSP universe. On average, the IIR signal is available for 3.74% of CRSP names, which on average make up 4.35% of total market capitalization.

4 Does IIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IIR portfolio and sells the low IIR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IIR strategy earns an average return of 0.23% per month with a t-statistic of 3.05. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.22% to 0.27% per month and have t-statistics exceeding 2.85 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.24, with a t-statistic of 4.51 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 372 stocks and an average market capitalization of at least \$1,079 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.32. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-29bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.16. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the

IIR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the IIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IIR, as well as average returns and alphas for long/short trading IIR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IIR strategy achieves an average return of 30 bps/month with a t-statistic of 2.74. Among these large cap stocks, the alphas for the IIR strategy relative to the five most common factor models range from 30 to 37 bps/month with t-statistics between 2.65 and 3.31.

5 How does IIR perform relative to the zoo?

Figure 2 puts the performance of IIR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IIR strategy falls in the distribution. The IIR strategy’s gross (net) Sharpe ratio of 0.43 (0.35) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IIR strategy (red line).² Ignoring trading costs, a \$1 invested in the IIR strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$2.62 which ranks the IIR strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IIR strategy would have yielded \$1.85 which ranks the IIR strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IIR relative to those. Panel A shows that the IIR strategy gross alphas fall between the 49 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases IIR ranks between the 67 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of IIR with 208 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IIR or at least to weaken the power IIR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IIR conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IIR}IIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IIR. Stocks are finally grouped into five IIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IIR trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IIR signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-0.65, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on IIR is -0.45.

Similarly, Table 5 reports results from spanning tests that regress returns to the IIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IIR strategy earns alphas that range from 21-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.71, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IIR trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.42.

7 Does IIR add relative to the whole zoo?

Finally, we can ask how much adding IIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the IIR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes IIR grows to \$1083.82.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IIR is available.

8 Conclusion

This study provides compelling evidence that Intangible Investment Restraint (IIR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IIR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on IIR delivers an annualized Sharpe ratio of 0.43 gross (0.35 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. More importantly, the strategy generates statistically significant abnormal returns of 22 basis points per month (t -statistic = 2.85) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 19 basis point monthly alpha (t -statistic = 2.42) after controlling for twelve factors, including six closely related financing and investment strategies, underscores that IIR captures unique information about future returns not subsumed by existing predictors.

These results have important practical implications for both investors and corporate managers. For investors, the IIR signal offers a viable strategy for enhancing portfolio returns, with the net Sharpe ratio of 0.35 suggesting profitability remains after realistic transaction costs. For corporate managers, our findings highlight the market’s valuation of intangible investment restraint, potentially informing capital allocation decisions and investment policies.

Despite these robust findings, several limitations merit consideration. First, while we control for numerous factors, the possibility remains that IIR proxies for an omitted risk factor or behavioral bias not captured in our analysis. Second, the implementation of IIR-based strategies may face practical challenges in real-world settings, including capacity constraints and market impact costs beyond those captured in our

net return calculations. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the IIR effect could provide deeper insights into why firms exhibiting intangible investment restraint generate superior returns. Second, examining the time-varying nature of the IIR premium and its relationship with macroeconomic conditions could enhance our understanding of when this strategy performs best. Third, extending the analysis to international markets would test the global applicability of our findings. Finally, exploring the interaction between IIR and firm characteristics such as industry classification, growth opportunities, and financial constraints could reveal important heterogeneity in the predictive power of this signal. Such investigations would not only validate the robustness of IIR as a return predictor but also contribute to our broader understanding of how intangible capital investments influence asset prices in modern economies.

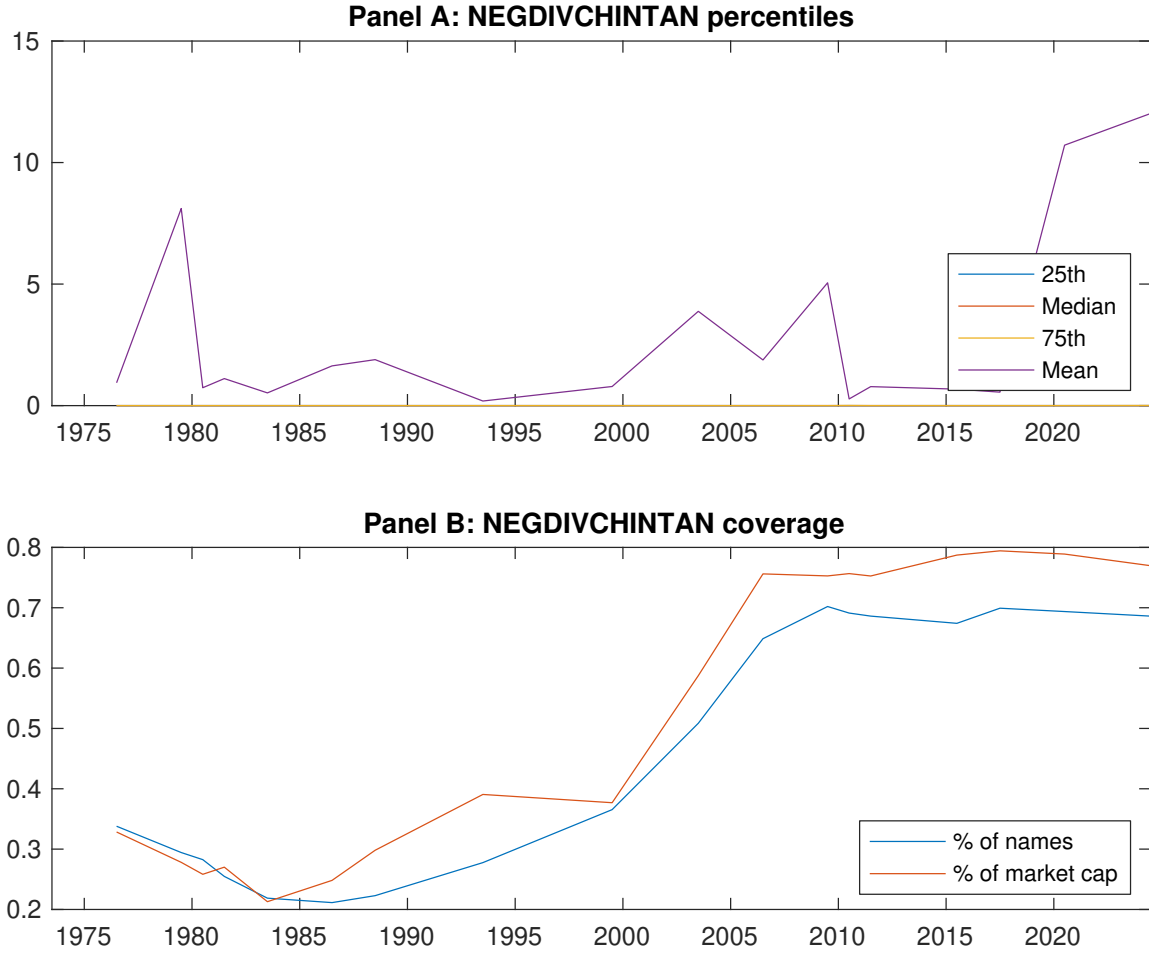


Figure 1: Times series of IIR percentiles and coverage.
This figure plots descriptive statistics for IIR. Panel A shows cross-sectional percentiles of IIR over the sample. Panel B plots the monthly coverage of IIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on IIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.72]	0.68 [3.42]	0.70 [3.46]	0.73 [3.76]	0.77 [3.95]	0.23 [3.05]
α_{CAPM}	-0.13 [-2.21]	0.02 [0.31]	0.04 [0.53]	0.09 [1.33]	0.11 [1.96]	0.24 [3.18]
α_{FF3}	-0.17 [-3.04]	0.00 [0.01]	0.05 [0.65]	0.06 [0.97]	0.09 [1.64]	0.27 [3.47]
α_{FF4}	-0.13 [-2.32]	0.01 [0.17]	0.05 [0.61]	0.08 [1.16]	0.12 [2.12]	0.26 [3.28]
α_{FF5}	-0.18 [-3.09]	-0.14 [-2.15]	-0.11 [-1.48]	-0.09 [-1.45]	0.05 [0.79]	0.22 [2.90]
α_{FF6}	-0.15 [-2.56]	-0.12 [-1.82]	-0.10 [-1.34]	-0.07 [-1.08]	0.08 [1.29]	0.22 [2.85]
Panel B: Fama and French (2018) 6-factor model loadings for IIR-sorted portfolios						
β_{MKT}	1.03 [77.24]	1.04 [68.54]	1.02 [58.99]	1.01 [72.00]	1.03 [75.75]	0.00 [0.02]
β_{SMB}	-0.01 [-0.49]	0.09 [3.91]	0.13 [4.84]	0.12 [5.64]	-0.05 [-2.37]	-0.04 [-1.42]
β_{HML}	0.14 [5.30]	-0.10 [-3.31]	-0.13 [-3.99]	-0.06 [-2.23]	-0.02 [-0.65]	-0.15 [-4.39]
β_{RMW}	0.08 [3.11]	0.24 [8.03]	0.36 [10.73]	0.32 [11.64]	0.04 [1.50]	-0.04 [-1.16]
β_{CMA}	-0.07 [-1.85]	0.26 [5.88]	0.14 [2.71]	0.17 [4.24]	0.17 [4.20]	0.24 [4.51]
β_{UMD}	-0.05 [-3.63]	-0.03 [-2.18]	-0.01 [-0.84]	-0.03 [-2.49]	-0.05 [-3.43]	0.00 [0.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	372	404	425	412	374	
me (\$10 ⁶)	2666	1134	1079	1111	2495	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23 [3.05]	0.24 [3.18]	0.27 [3.47]	0.26 [3.28]	0.22 [2.90]	0.22 [2.85]
Quintile	NYSE	EW	0.13 [2.32]	0.13 [2.33]	0.12 [2.12]	0.13 [2.14]	0.08 [1.46]	0.09 [1.60]
Quintile	Name	VW	0.20 [2.70]	0.21 [2.84]	0.23 [3.09]	0.22 [2.87]	0.19 [2.52]	0.19 [2.45]
Quintile	Cap	VW	0.33 [3.36]	0.35 [3.51]	0.39 [3.97]	0.39 [3.86]	0.32 [3.20]	0.33 [3.21]
Decile	NYSE	VW	0.25 [2.20]	0.26 [2.28]	0.30 [2.58]	0.33 [2.79]	0.30 [2.48]	0.32 [2.69]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.19 [2.54]	0.20 [2.66]	0.23 [2.97]	0.22 [2.91]	0.20 [2.57]	0.19 [2.49]
Quintile	NYSE	EW	-0.08 [-1.16]					
Quintile	Name	VW	0.16 [2.17]	0.17 [2.33]	0.19 [2.60]	0.19 [2.53]	0.16 [2.19]	0.15 [2.09]
Quintile	Cap	VW	0.29 [2.96]	0.30 [2.98]	0.34 [3.40]	0.34 [3.38]	0.29 [2.83]	0.28 [2.78]
Decile	NYSE	VW	0.21 [1.84]	0.20 [1.72]	0.23 [2.04]	0.25 [2.21]	0.23 [1.99]	0.24 [2.07]

Table 3: Conditional sort on size and IIR

This table presents results for conditional double sorts on size and IIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IIR and short stocks with low IIR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IIR Quintiles					IIR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.78 [2.95]	0.93 [3.14]	0.81 [2.89]	0.88 [3.18]	0.90 [3.30]	0.11 [1.14]	0.09 [0.94]	0.10 [0.95]	0.10 [0.94]	0.09 [0.87]	0.09 [0.90]
	(2)	0.88 [3.32]	0.86 [3.29]	0.88 [3.43]	0.84 [3.32]	0.89 [3.48]	0.01 [0.09]	0.02 [0.21]	0.03 [0.27]	0.01 [0.05]	-0.06 [-0.54]	-0.07 [-0.62]
	(3)	0.72 [2.96]	0.92 [3.93]	0.81 [3.42]	0.90 [3.72]	0.80 [3.37]	0.08 [0.82]	0.10 [0.98]	0.09 [0.93]	0.03 [0.31]	0.02 [0.17]	-0.02 [-0.22]
	(4)	0.84 [3.72]	0.79 [3.50]	0.81 [3.52]	0.81 [3.65]	0.77 [3.44]	-0.08 [-0.78]	-0.06 [-0.61]	-0.08 [-0.84]	-0.04 [-0.38]	-0.17 [-1.68]	-0.13 [-1.25]
	(5)	0.44 [2.13]	0.63 [3.25]	0.60 [3.04]	0.70 [3.79]	0.74 [3.68]	0.30 [2.74]	0.32 [2.94]	0.35 [3.21]	0.37 [3.31]	0.30 [2.65]	0.32 [2.81]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IIR Quintiles					IIR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	207	205	205	206	206	26	23	22	22	25	
	(2)	67	67	67	67	66	47	47	46	45	47	
	(3)	48	48	48	48	48	84	86	84	83	84	
	(4)	40	40	40	40	40	182	184	178	182	183	
(5)	37	37	37	37	37	1619	1259	1149	1153	1624		

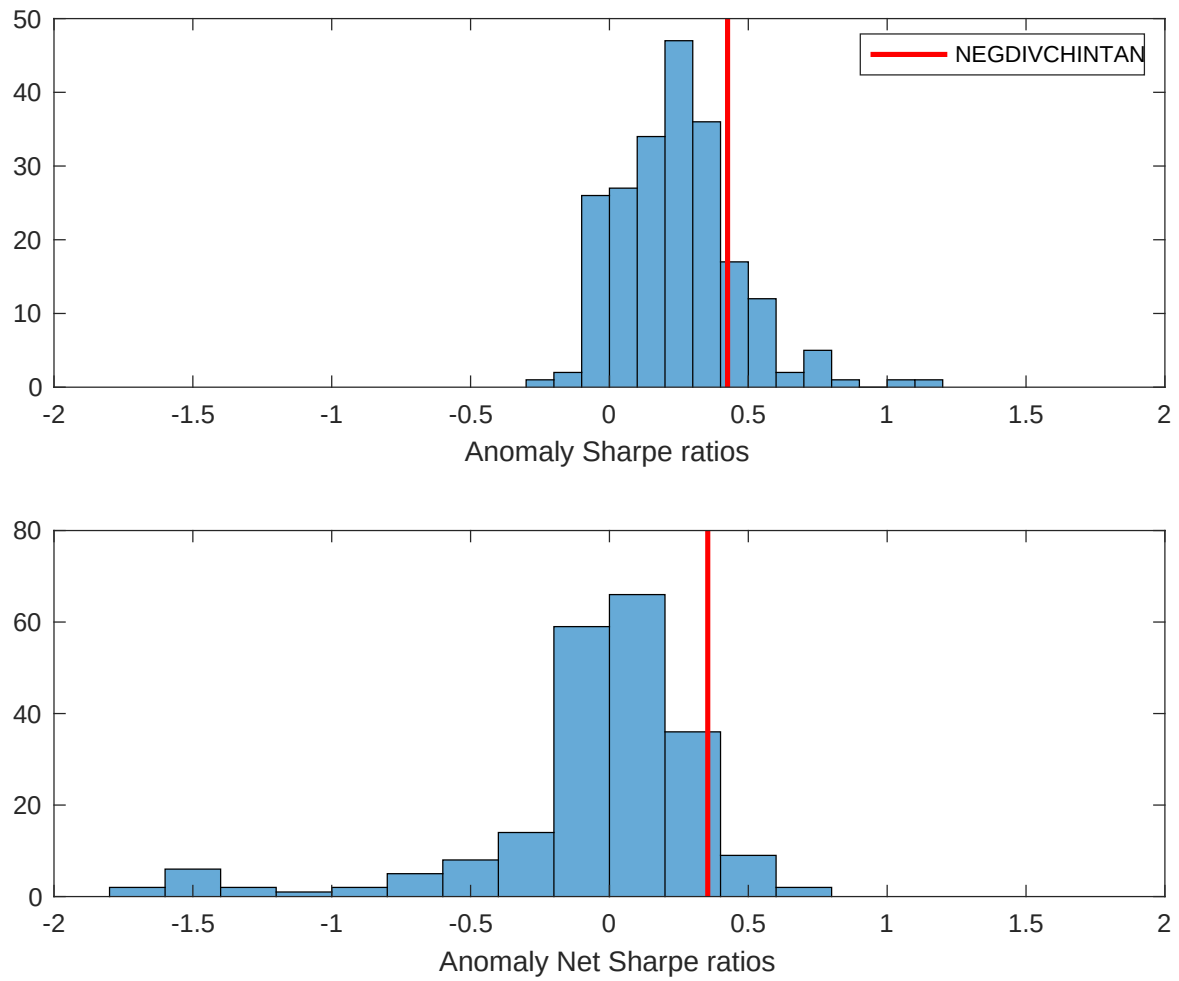


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

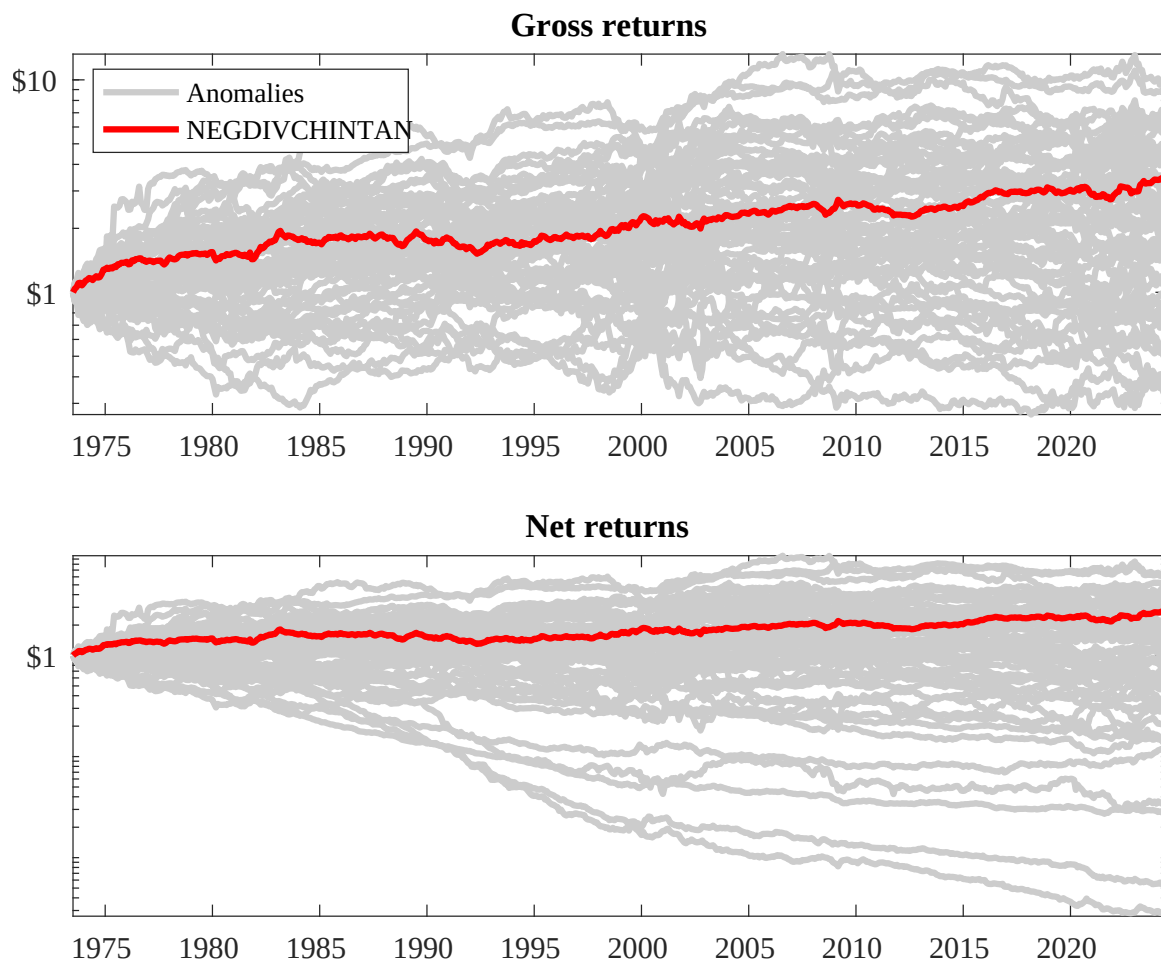
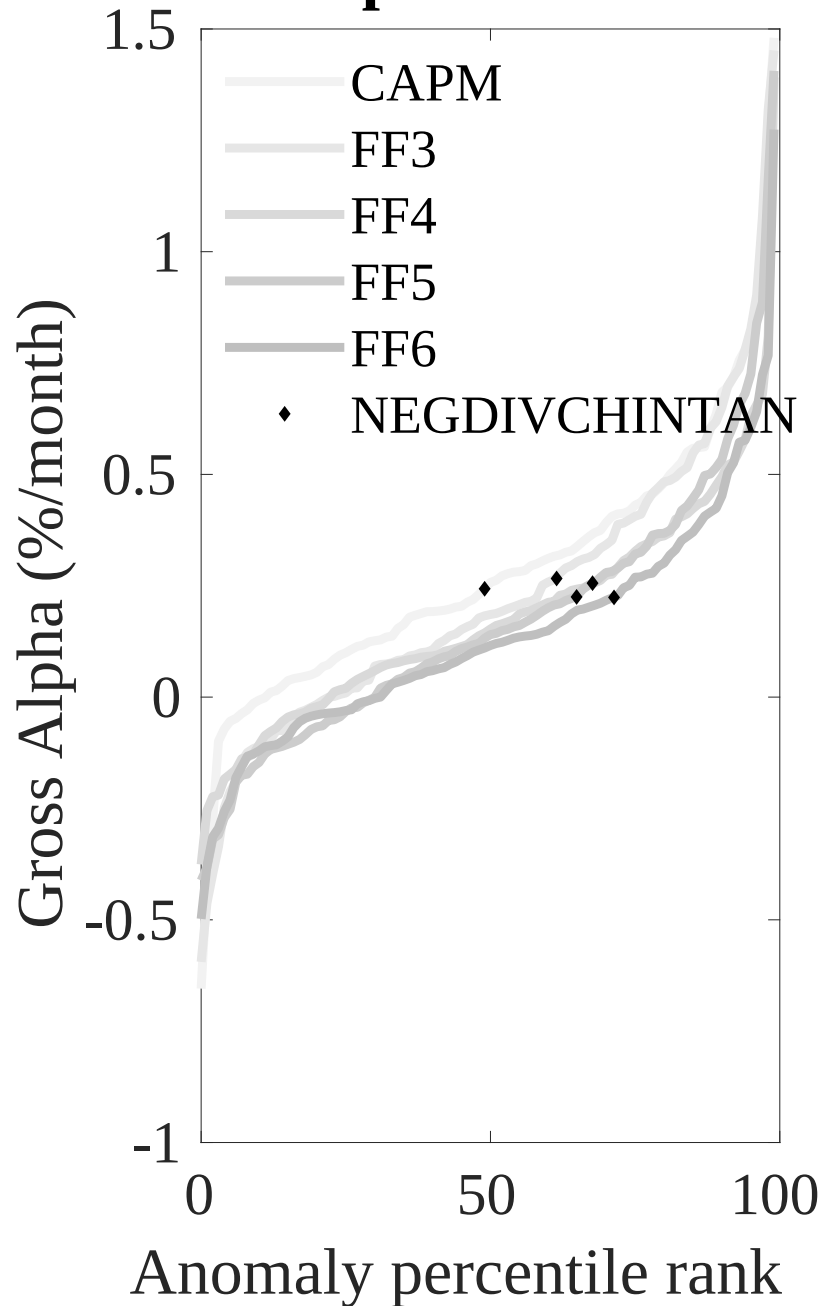


Figure 3: Dollar invested.

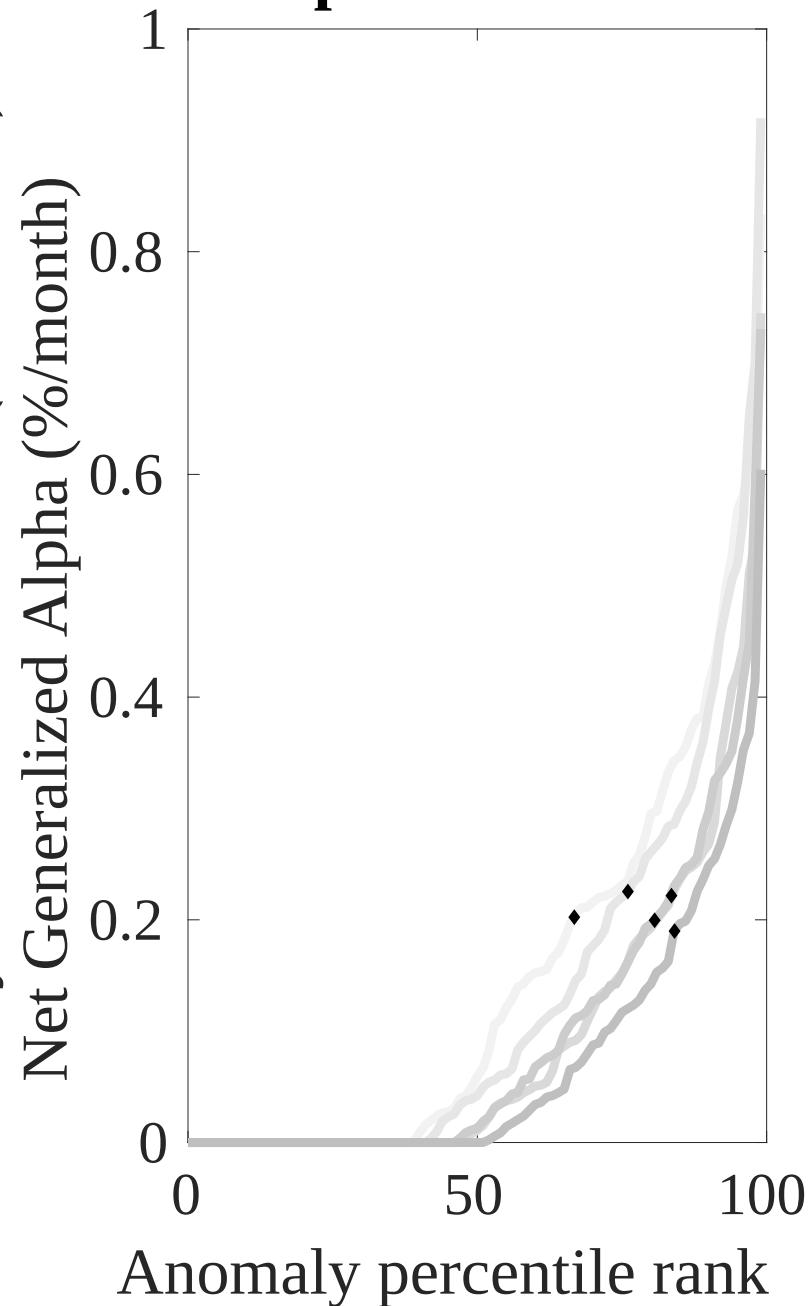
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



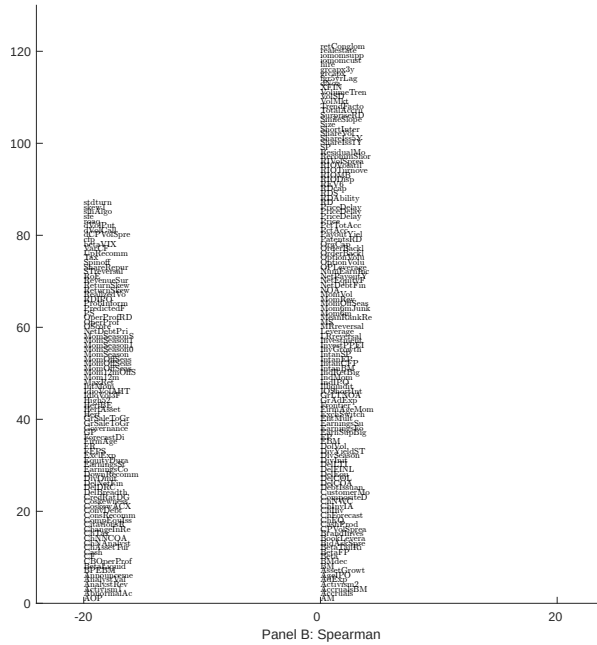
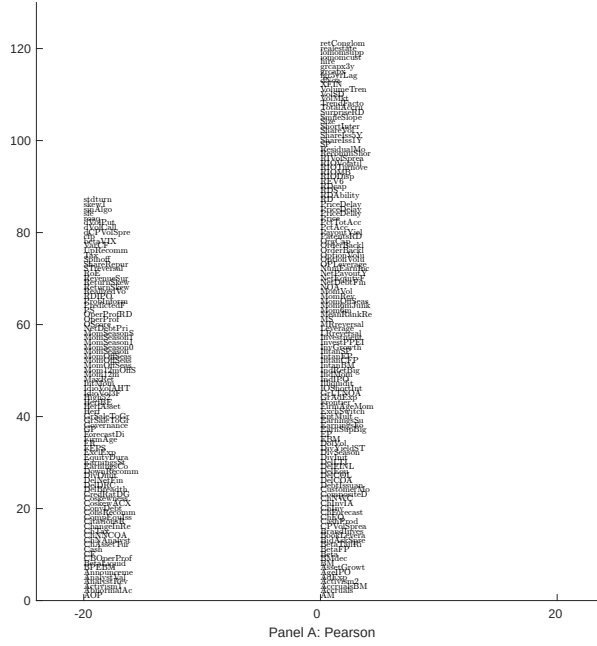


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 208 filtered anomaly signals with IIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

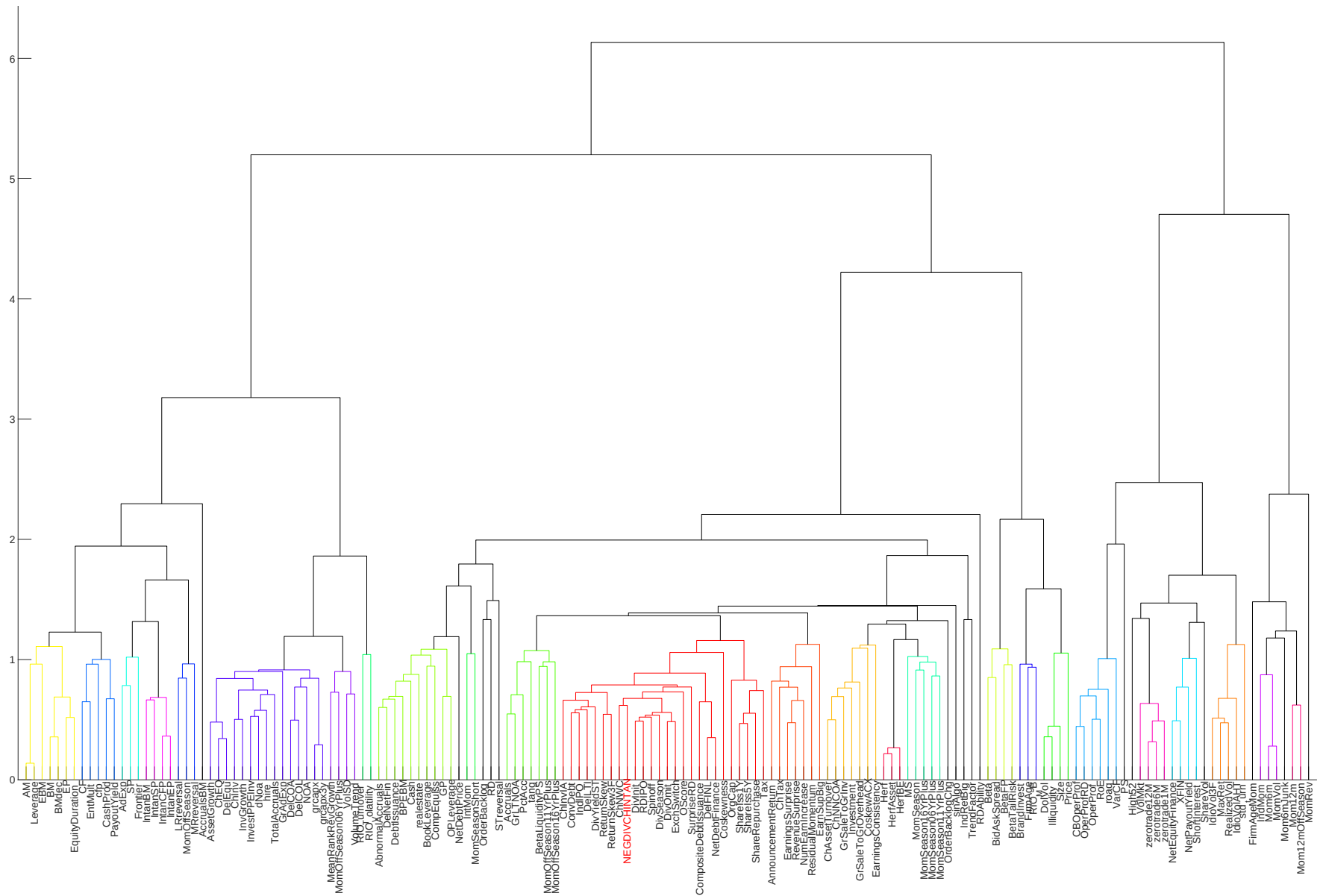


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

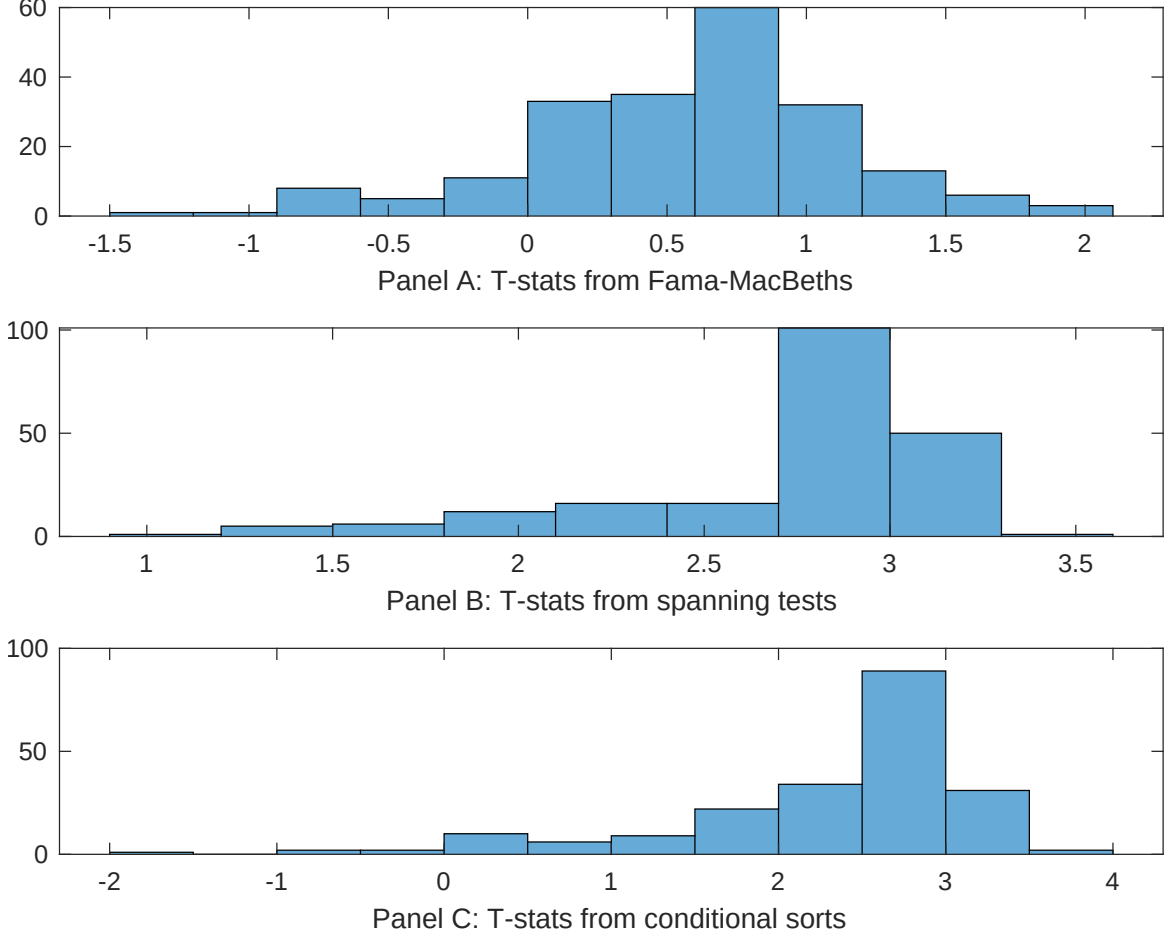


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IIR conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IIR}IIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IIR. Stocks are finally grouped into five IIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IIR trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IIR}IIR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.52]	0.13 [5.28]	0.13 [5.28]	0.13 [5.18]	0.14 [5.70]	0.12 [4.96]	0.14 [5.51]
IIR	0.17 [0.26]	0.26 [0.41]	0.64 [0.11]	0.16 [0.25]	-0.38 [-0.65]	0.32 [0.51]	-0.28 [-0.45]
Anomaly 1	0.21 [6.91]						0.13 [1.98]
Anomaly 2		0.20 [8.16]					-0.20 [-0.29]
Anomaly 3			0.15 [3.86]				-0.28 [-0.54]
Anomaly 4				0.45 [1.86]			0.13 [0.60]
Anomaly 5					0.99 [8.00]		0.69 [5.16]
Anomaly 6						0.17 [4.74]	0.75 [2.37]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	1	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IIR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [2.71]	0.21 [2.78]	0.22 [2.87]	0.22 [2.81]	0.22 [2.87]	0.21 [2.72]	0.19 [2.42]
Anomaly 1	11.98 [3.09]						9.10 [2.19]
Anomaly 2		11.43 [2.70]					9.52 [2.14]
Anomaly 3			9.23 [2.41]				10.02 [2.04]
Anomaly 4				4.54 [1.29]			1.86 [0.47]
Anomaly 5					6.63 [1.47]		-2.70 [-0.52]
Anomaly 6						3.98 [1.29]	4.67 [1.50]
mkt	1.59 [0.86]	-0.03 [-0.02]	-0.13 [-0.07]	-0.20 [-0.11]	-0.06 [-0.03]	0.24 [0.14]	1.46 [0.78]
smb	-1.31 [-0.44]	-6.17 [-2.27]	-5.12 [-1.90]	-4.97 [-1.83]	-5.64 [-2.07]	-4.40 [-1.59]	-1.89 [-0.59]
hml	-15.71 [-4.55]	-16.57 [-4.82]	-18.27 [-5.25]	-16.83 [-4.87]	-17.34 [-5.02]	-16.02 [-4.53]	-15.64 [-4.37]
rmw	-10.97 [-2.62]	-5.07 [-1.44]	-2.98 [-0.85]	-3.01 [-0.85]	-3.79 [-1.08]	-2.50 [-0.69]	-7.54 [-1.73]
cma	17.55 [3.03]	22.39 [4.20]	16.17 [2.50]	22.72 [4.04]	17.33 [2.29]	24.07 [4.51]	7.71 [0.98]
umd	-0.24 [-0.14]	-0.90 [-0.50]	-0.09 [-0.05]	-0.02 [-0.01]	-0.02 [-0.01]	-0.46 [-0.26]	-1.13 [-0.63]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	6	6	6	5	5	5	8

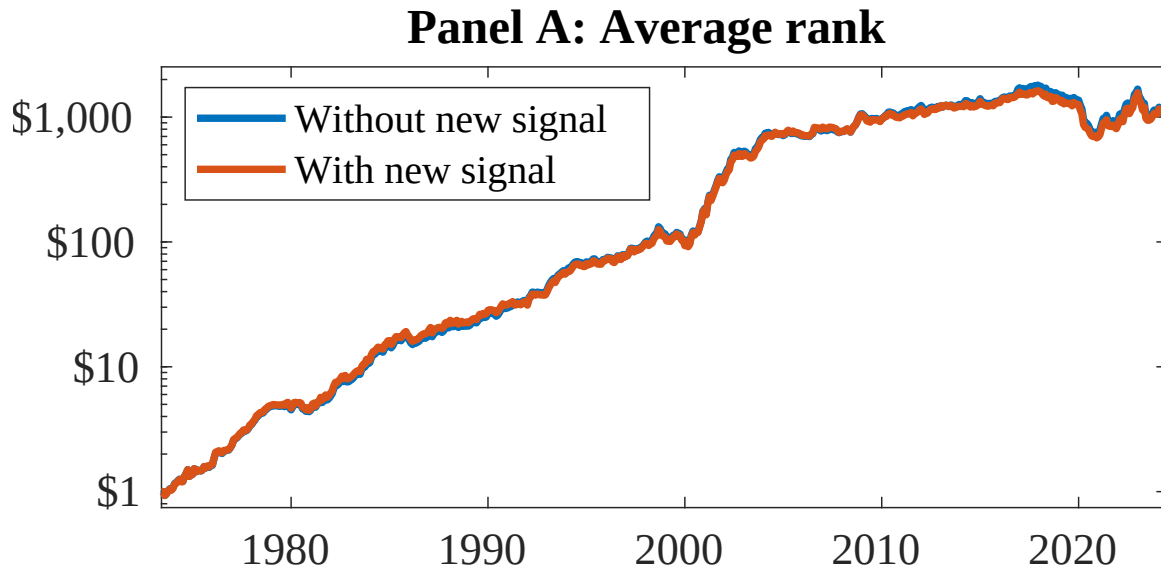


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as IIR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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